Univariate Anomaly & Pattern Detection

Objectives

- Adopt a score-first time-series anomaly detection (TSAD) mindset: score first, threshold second, alarms last.
- Core methods: z-score, MAD, STL residuals, CUSUM/PELT, Matrix Profile, DTW.
- Define evaluation outputs: range-aware precision/recall/F1, `Precision@k`, latency.
- Compare methods across datasets/metrics.

Agenda

- Definitions and process-centric TSAD framing.
- Anomaly taxonomy and visual grounding.
- Evaluation design: event-aware and threshold-aware reporting.
- Thresholding and EVT/POT tails.
- Methods: z-score/MAD, STL residuals, CUSUM/PELT, Matrix Profile, DTW.
- Labs.

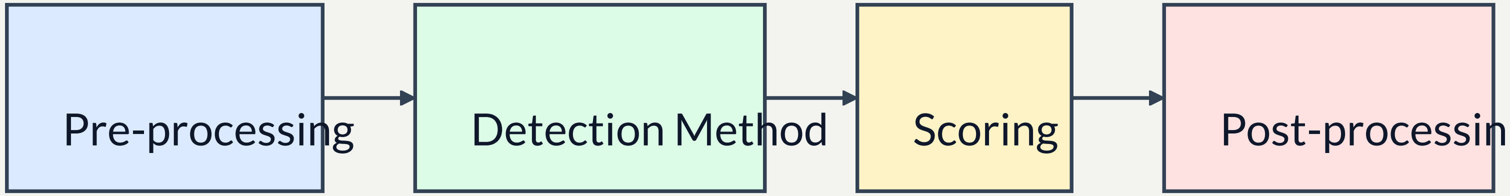
Framing: Score-First TSAD

- **Formal notation**: univariate series $T \in R^n$; subsequence $T_{(}i, l) = [T_i, \ldots, T_{(i+l-1)}]$ with $l << n$.
- **Anomaly score**: real-valued signal of abnormality per timestamp or window.
- **Threshold**: decision cutoff turning scores into alarms (`score > threshold`).

Framing: Alarms

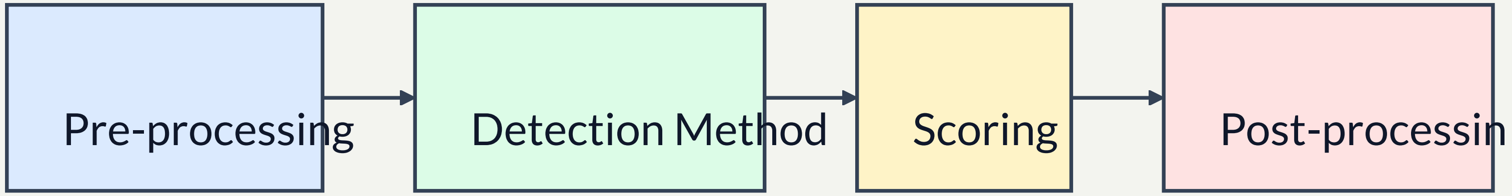
- **Binary alarm**: yes/no indicator at each timestamp after thresholding.
- **Alarm window/event**: contiguous span of alarmed timestamps, used for operations and range-aware evaluation.

- **Process-centric pipeline**:



- **Windowing reality**: most TSAD pipelines begin by converting time series into a sliding-window matrix.
- Rigour checklist per method: state assumptions, score construction, threshold policy, and post-processing.
- Why this framing: it keeps detectors comparable even when methods differ.

Visual: Process-Centric Pipeline



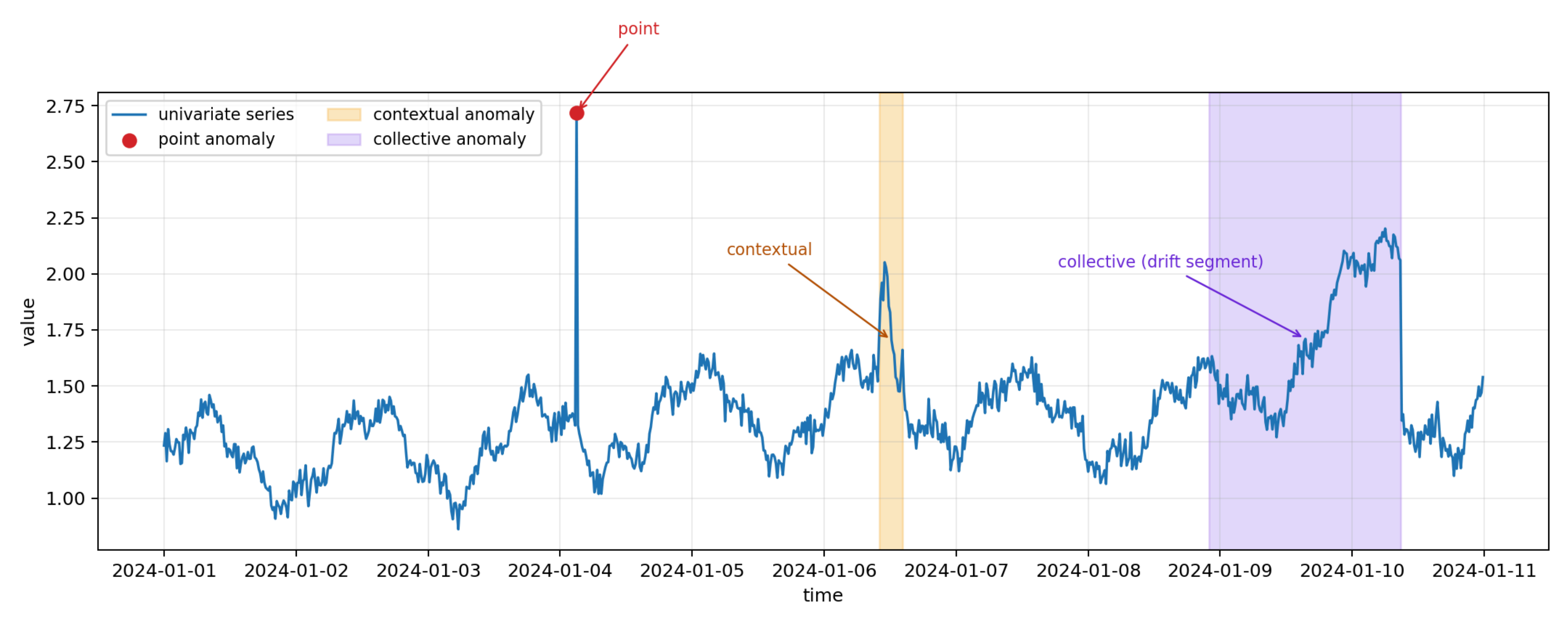
Anomaly Types (Definitions)

- **Point anomaly**: isolated timestamp with unlikely value (example: one sudden CPU spike).
- **Contextual anomaly**: value is abnormal only in context (example: nighttime CPU level that is normal at noon).
- **Collective anomaly**: a segment pattern is abnormal as a whole (example: drift, sustained flatline, regime shift).

- Assumption: point + contextual are treated as point-based anomalies; collective anomalies are treated as sequence-based.
- Important caveat: a sequence can be anomalous even when all values stay within the global range; the anomaly is in temporal shape/context.
- Edge case: whole-sequence anomaly detection treats an entire series as the anomaly unit.

- Today:
  - Point: rolling z-score, MAD.
  - Contextual: STL + residual thresholding.
  - Collective: change points, Matrix Profile discords, DTW template distance.

Visual: Anomaly Type Sketch



Evaluation Preliminaries

- Point-based view: evaluate per timestamp with true labels y_t and predictions $\hat{y_t}$ (TP/FP/TN/FN).
- Range-aware view: contiguous positives are grouped into events/ranges before evaluation.
- Ground truth is represented as anomaly ranges/events R_{true}, not only isolated points.
- Detector output is converted into predicted alarm ranges/events R_{pred} after thresholding and post-processing.

- Overlap rule: a predicted range counts as a hit if it overlaps at least one true range.
- Optional stricter scoring: assign partial credit based on overlap amount instead of binary hit/miss.

Evaluation Concepts

- **Point-wise precision**: fraction of predicted anomalous timestamps that are truly anomalous.
- **Point-wise recall**: fraction of true anomalous timestamps that are detected.
- **Point-wise F1**: harmonic mean of point-wise precision/recall.
- `Precision@k`: among top-k highest scores, how many timestamps fall in true anomaly ranges.
- Classic point-adjusted scoring can overestimate performance if one hit credits an entire anomaly segment.

- **Range-aware precision**: fraction of predicted alarm ranges that overlap true anomaly ranges, with optional partial-overlap credit.
- **Range-aware recall**: fraction of true anomaly ranges covered by predicted alarm ranges, often with early-hit preference.
- **Range-aware F1**: harmonic mean of range-aware precision/recall.

- **Latency-to-first-detection**: delay from anomaly start to first alarm inside each true range.
- **NAB-style latency reward**: earlier valid detections get higher reward; false alarms outside windows are penalized.

Evaluation Formula Snapshot

- Point-wise confusion matrix terms: TP, FP, TN, FN.
- Precision:
  - $$\text{Precision} = \frac{TP}{TP + FP}$$
- Recall:
  - $$\text{Recall} = \frac{TP}{TP + FN}$$

- F1:
  - $$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$
- Precision@k (ranking view):
  - $$\text{Precision@k} = \frac{1}{k} \sum_{t \in \text{TopK}(s)} \mathbf{1}\{t \in \square_{true}\}$$

- Range-aware precision (binary overlap):
  - $$\text{Precision}_{\text{range}} = \frac{1}{|R_{\text{pred}}|} \sum_{r \in R_{\text{pred}}} \mathbf{1}\{\exists g \in R_{\text{true}} : |r \cap g| > 0\}$$
- Range-aware recall (binary overlap):
  - $$\text{Recall}_{\text{range}} = \frac{1}{|R_{\text{true}}|} \sum_{g \in R_{\text{true}}} \mathbf{1}\left\{\exists r \in R_{\text{pred}} : |r \cap g| > 0\right\}$$
- Threshold-free reporting:
  - Prefer AUC-PR / AUC-ROC / Range-AUC when score scaling differs between methods.

Threshold-Free Curves

- These metrics evaluate ranking quality over all thresholds instead of one fixed cutoff.
- **AUC-PR** is often the primary choice under strong class imbalance.
- **AUC-ROC** is useful but can appear optimistic when anomalies are rare.
- **Range-AUC** extends threshold-free evaluation to event/range detection.

AUC-PR Formula

- PR curve from threshold sweep:
  - $$\text{AUC-PR} = \int_0^1 \text{Precision}(r)\, dr$$
- Discrete approximation:
  - $$\text{AUC-PR} \approx \sum_{i=1}^{m} (R_i - R_{i-1})\, P_i$$
  - with recall/precision points (R_i, P_i) from sorted scores.

How to Read PR Curves

- Axes: x = Recall, y = Precision.
- Each point is one threshold; moving along the curve means changing the cutoff.
- Top-right is best (high precision and high recall at the same time).
- For imbalanced anomaly labels, PR is usually more diagnostic than ROC.
- Operational thresholding:
  - false alarms expensive $\longrightarrow$ choose higher-precision region.
  - misses expensive $\longrightarrow$ choose higher-recall region.

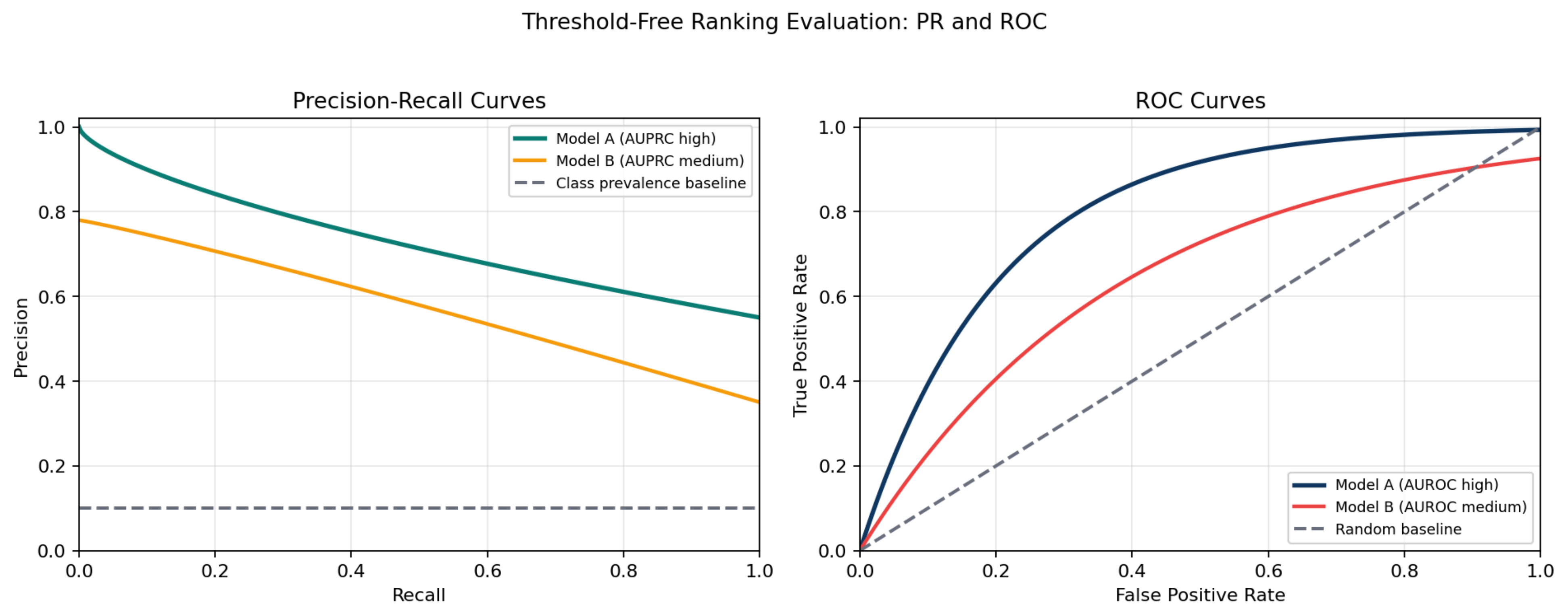
Class Prevalence Baseline

- In PR space, random ranking yields precision near class prevalence:
  - $$\pi = \frac{N_+}{N}$$
  - where N_+ is number of positives and N total samples.
- Interpretation:
  - PR curve above π indicates useful ranking signal.
  - If anomaly prevalence is 1%, random baseline precision is about 0.01.

AUC-ROC Formula

- ROC curve coordinates:
  - $$\text{TPR} = \frac{TP}{TP + FN}, \qquad \text{FPR} = \frac{FP}{FP + TN}$$
- Area under ROC:
  - $$\text{AUC-ROC} = \int_0^1 \text{TPR}(u)\, d\text{FPR}(u)$$
- Discrete approximation (threshold points sorted by increasing FPR):
  - $$\text{AUC-ROC} \approx \sum_{i=1}^{m} \frac{\text{TPR}_i + \text{TPR}_{i-1}}{2} \left(\text{FPR}_i - \text{FPR}_{i-1}\right)$$

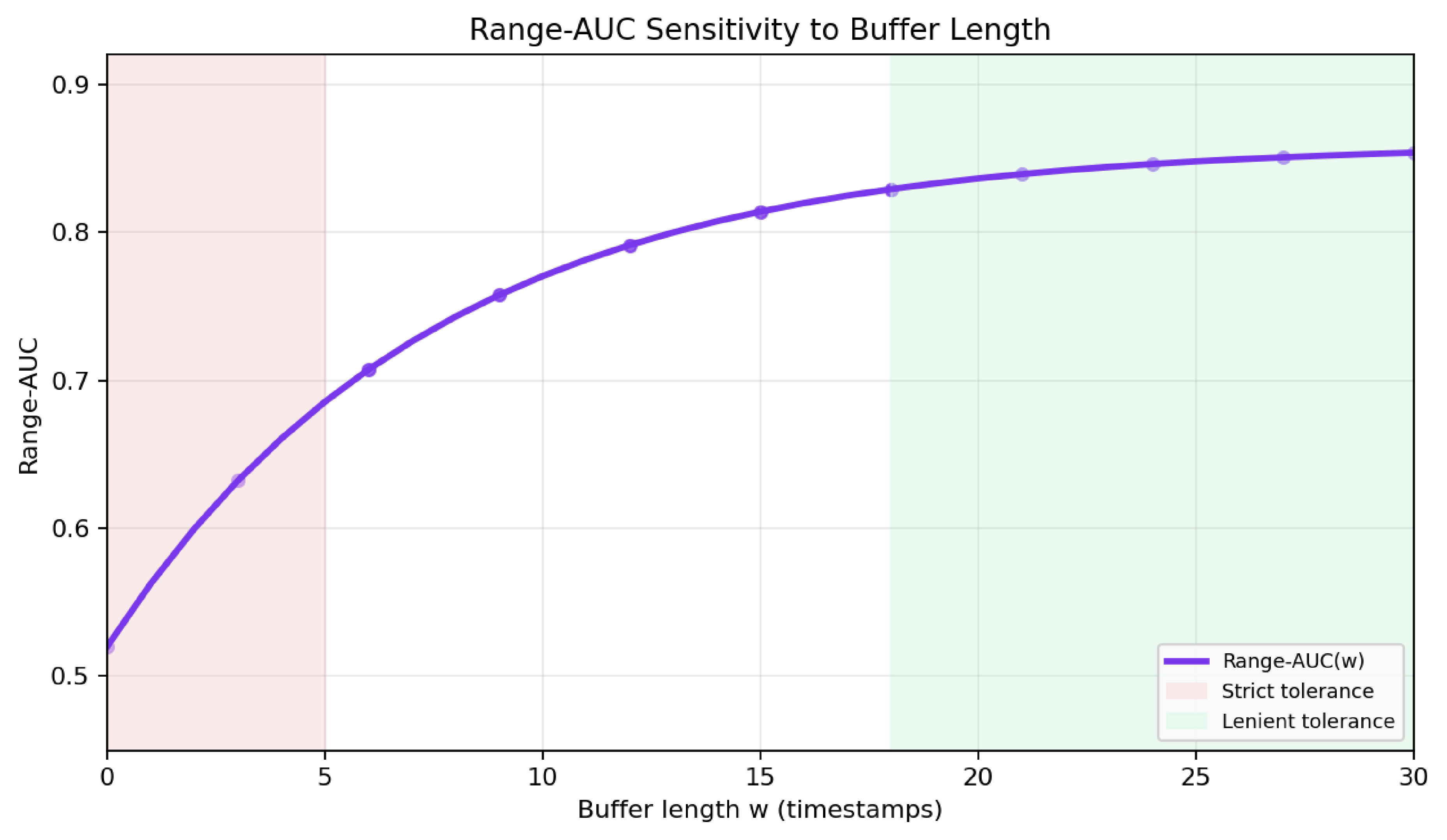
Visual: PR and ROC



Range-AUC Formula

- Integrate range-aware precision over range-aware recall:
  - $$\text{Range-AUC} = \int_0^1 \text{Range-Precision}(r)\, dr$$
- If tolerance buffer w varies, report average across a set $\square$:
  - $$\text{Range-AUC}_{\square} = \frac{1}{|\square|} \sum_{w \in \square} \text{Range-AUC}(w)$$

Visual: Range-AUC vs Buffer



Thresholding Definitions

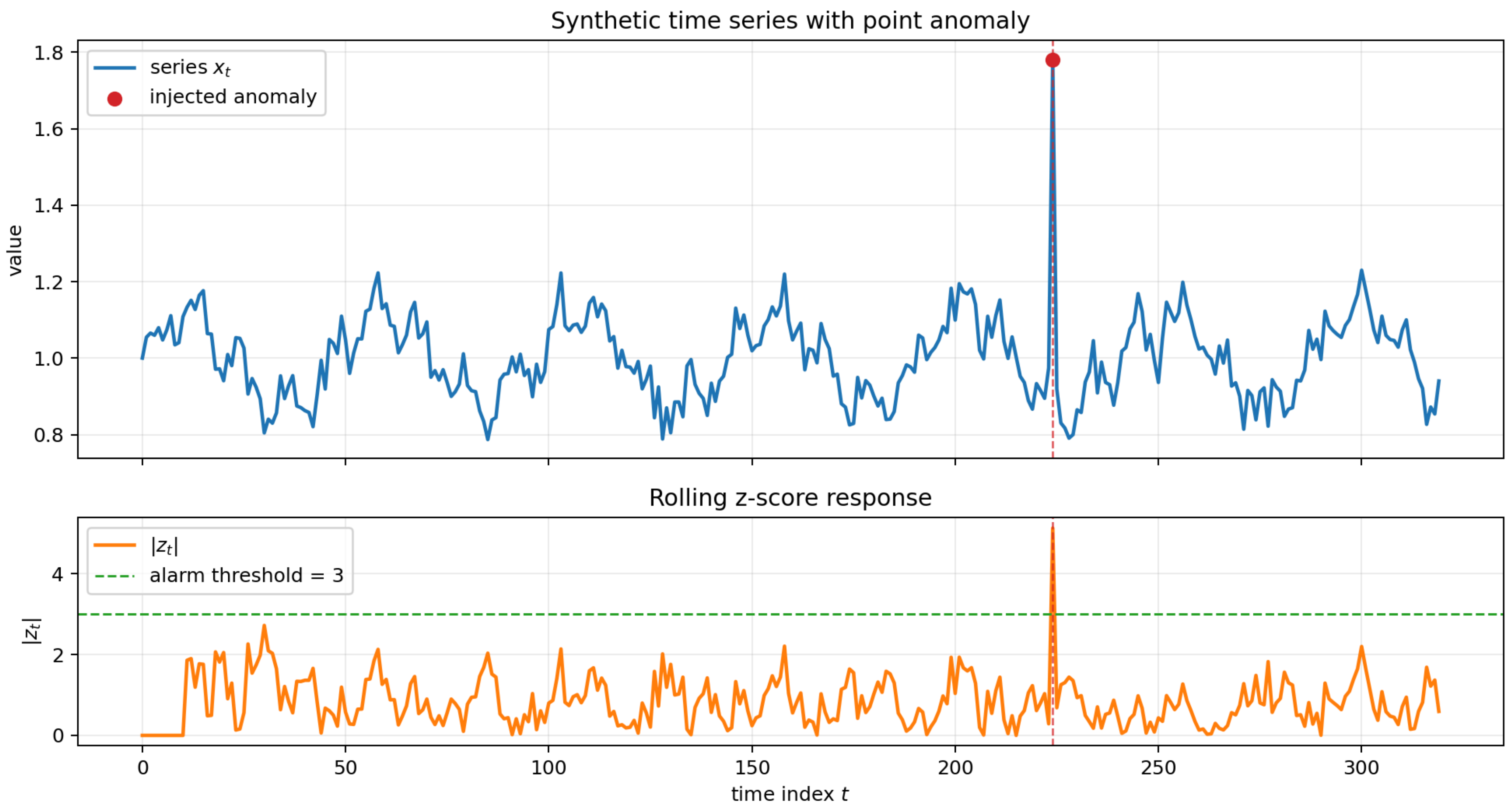
- **Static threshold**: single fixed cutoff for the full series.
- **Rolling threshold**: cutoff adapts with local rolling statistics (mean/std or median/MAD).
- **Seasonal quantile threshold**: cutoff conditioned on season bins (for example, hour-of-day).

- **Validation-based threshold selection**: choose threshold on validation data to match alert budget and miss-rate targets.
- **Event post-processing**: merge nearby alarms and apply minimum-duration rules before event-level scoring.
- Baseline $\mu + 3\sigma$ thresholding is simple but sensitive to extremes and cross-method score scaling differences.

- Dynamic threshold alternatives: adaptive thresholding, non-parametric dynamic thresholding, and POT-based tails.
- Robustness caveat: thresholded metrics can change sharply with score noise and small detection lag.
- Operational rule: threshold choice often matters as much as detector choice.

Classical Detectors I: z-score and MAD

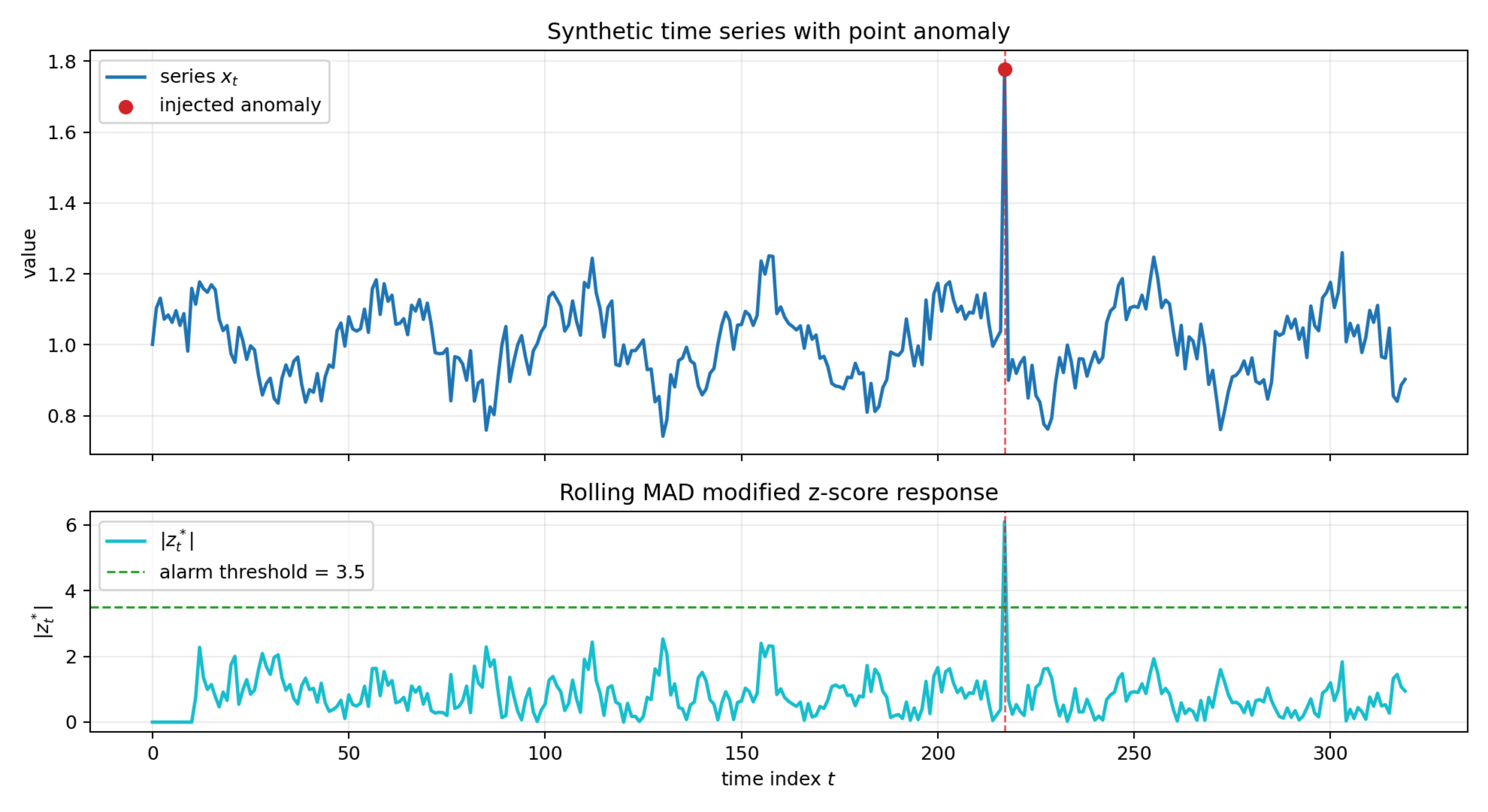
- **z-score (rolling)**:
  - Definition: $z_t = (x_t - \mu_t)/(\sigma_t + \varepsilon)$.
  - Alarm idea: large $|z_t|$ indicates outlier behavior.



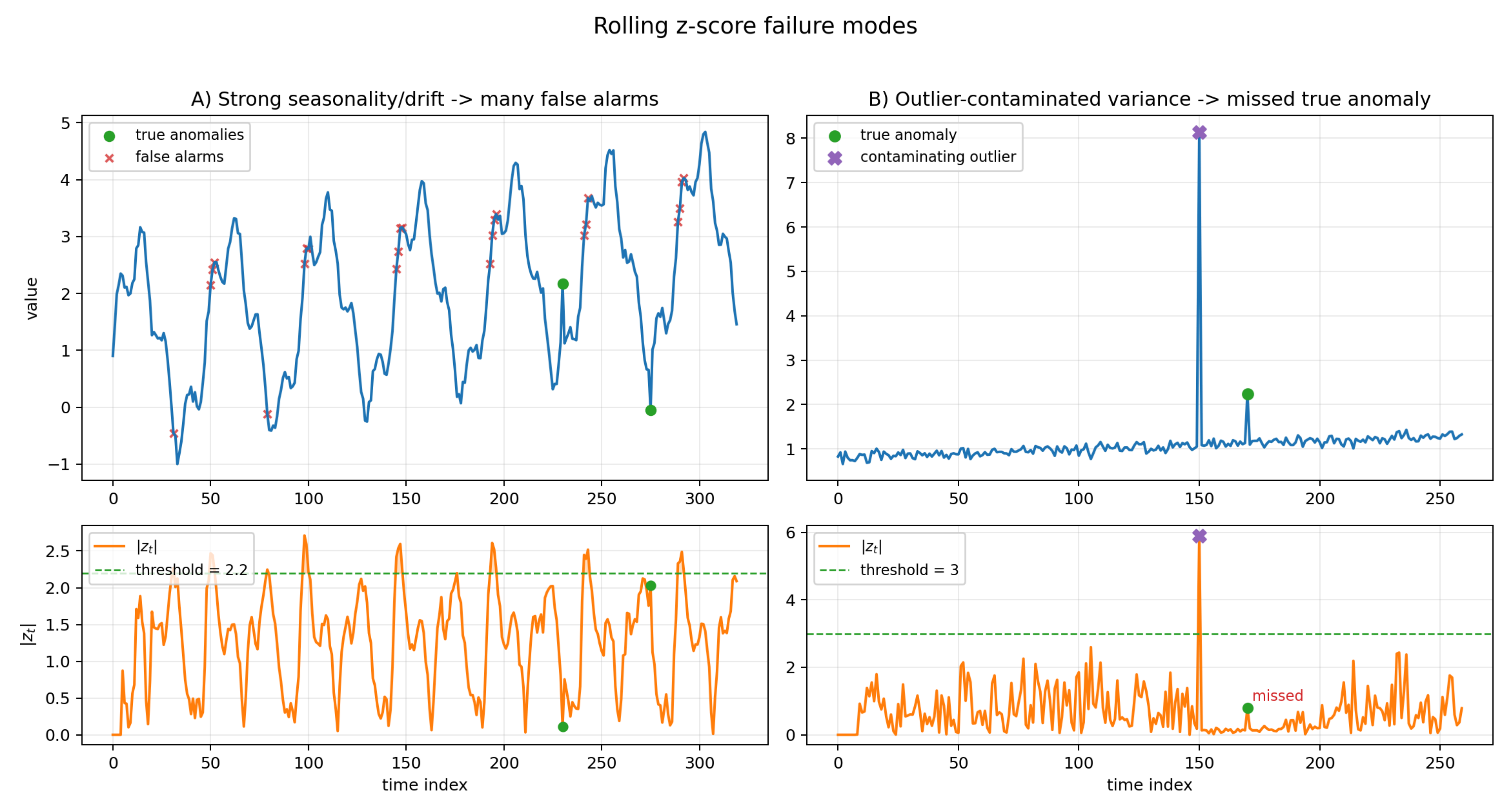
Synthetic time series with point anomaly
series x_t
injected anomaly
value
1.8
1.6
1.4
1.2
1.0
0.8
Rolling z-score response
$|z_t|$
alarm threshold = 3
$|z_t|$
4
2
0
0
50
100
150
200
250
300
time index t

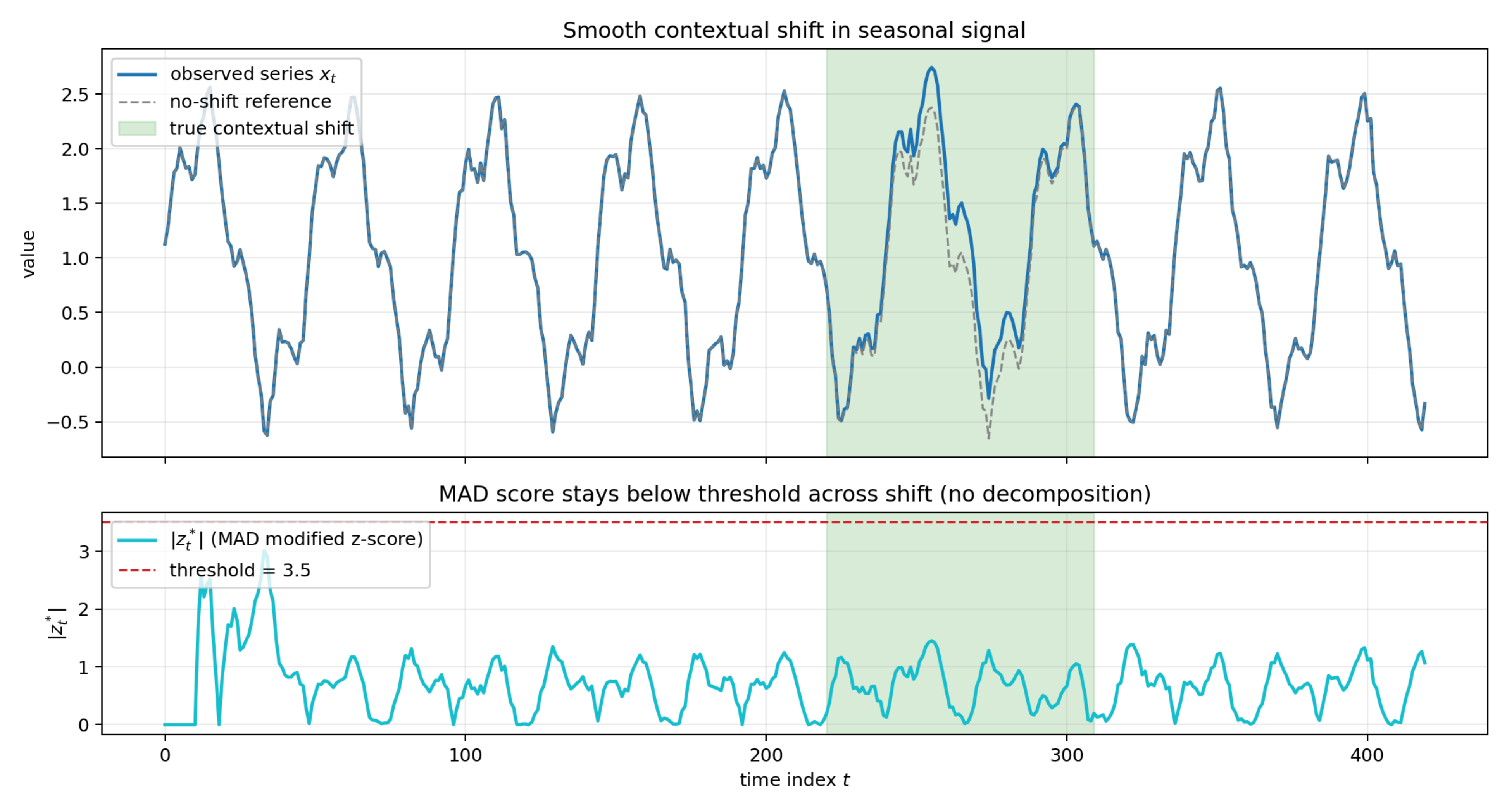
- **MAD (Median Absolute Deviation)**:
  - $MAD := median(|x - median(x)|)$.
  - **modified z-score**:

$$z_t^* = 0.6745(x_t - median_t)/(MAD_t + eps).$$


Synthetic time series with point anomaly
series x_t
injected anomaly
value
1.8
1.6
1.4
1.2
1.0
0.8
Rolling MAD modified z-score response
$|z_t^*|$
alarm threshold = 3.5
$|z_t^*|$
6
4
2
0
0
50
100
150
200
250
300
time index t

- Assumptions: these methods work best when local windows are approximately stationary.
- Implementation sensitivity: window length and warm-up (`min_periods`) **significantly** change false-positive rates.
- Failure modes:
  - z-score fails under strong seasonality/drift and outlier-contaminated variance.
  - MAD is robust but can still miss smooth contextual shifts without decomposition.


Rolling z-score failure modes
A) Strong seasonality/drift -> many false alarms
B) Outlier-contaminated variance -> missed true anomaly
true anomalies
false alarms
true anomaly
contaminating outlier
value
$|z_t|$
threshold = 2.2
threshold = 3
missed
time index
time index

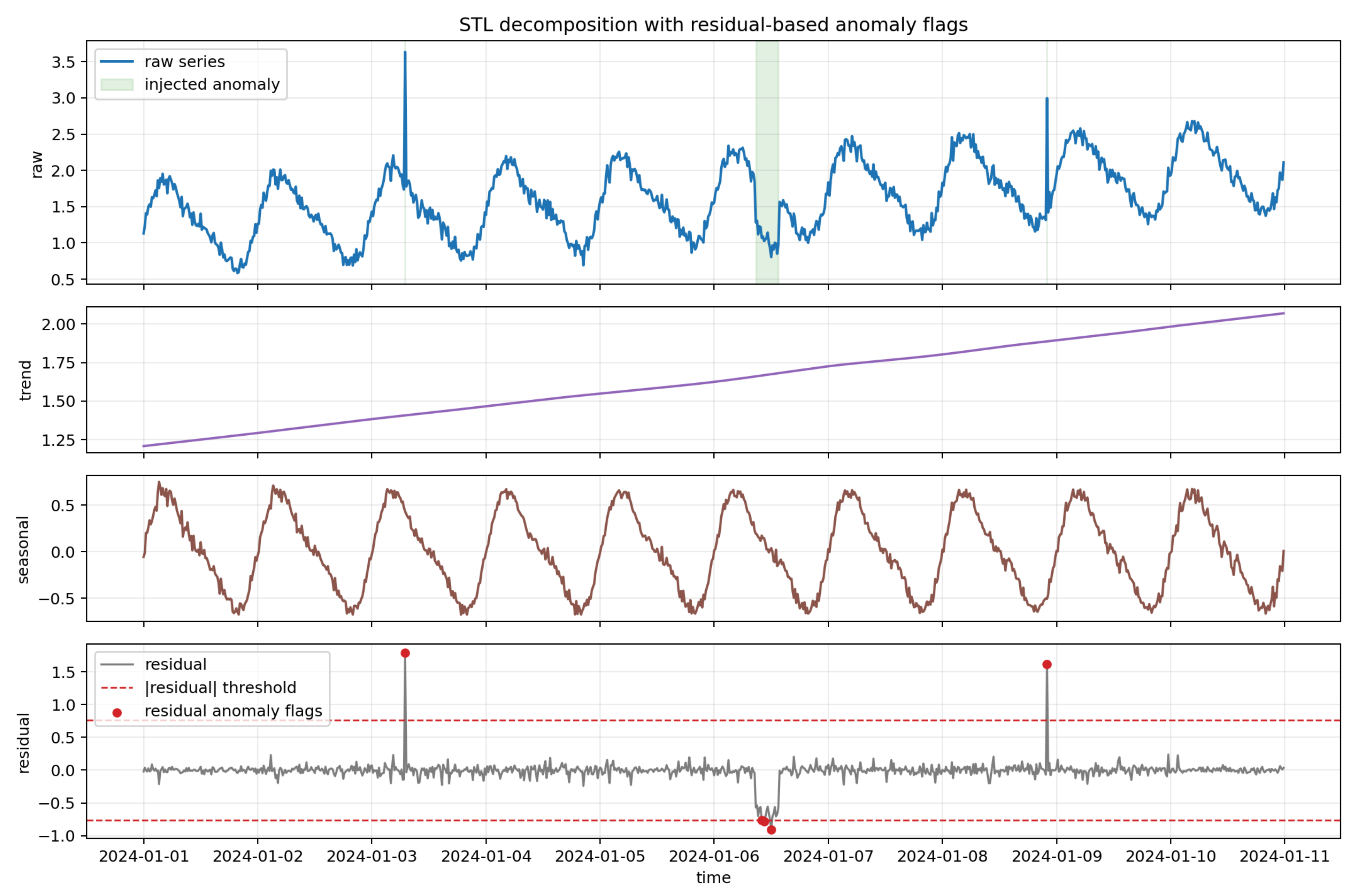

Smooth contextual shift in seasonal signal
observed series x_t
no-shift reference
true contextual shift
value
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
MAD score stays below threshold across shift (no decomposition)
$|z_t^*|$ (MAD modified z-score)
threshold = 3.5
$|z_t^*|$
3
2
1
0
0
100
200
300
400
time index t

Classical Detectors II: STL and Residual Scoring

- **STL** (Seasonal-Trend decomposition using LOESS):
  - Decomposition: $x_t = trend_t + seasonal_t + residual_t$.
- **Residual**: unexplained component after trend/seasonal removal.
  - Residual score example: $score_t = |residual_t|/(std(residual) + \varepsilon)$.

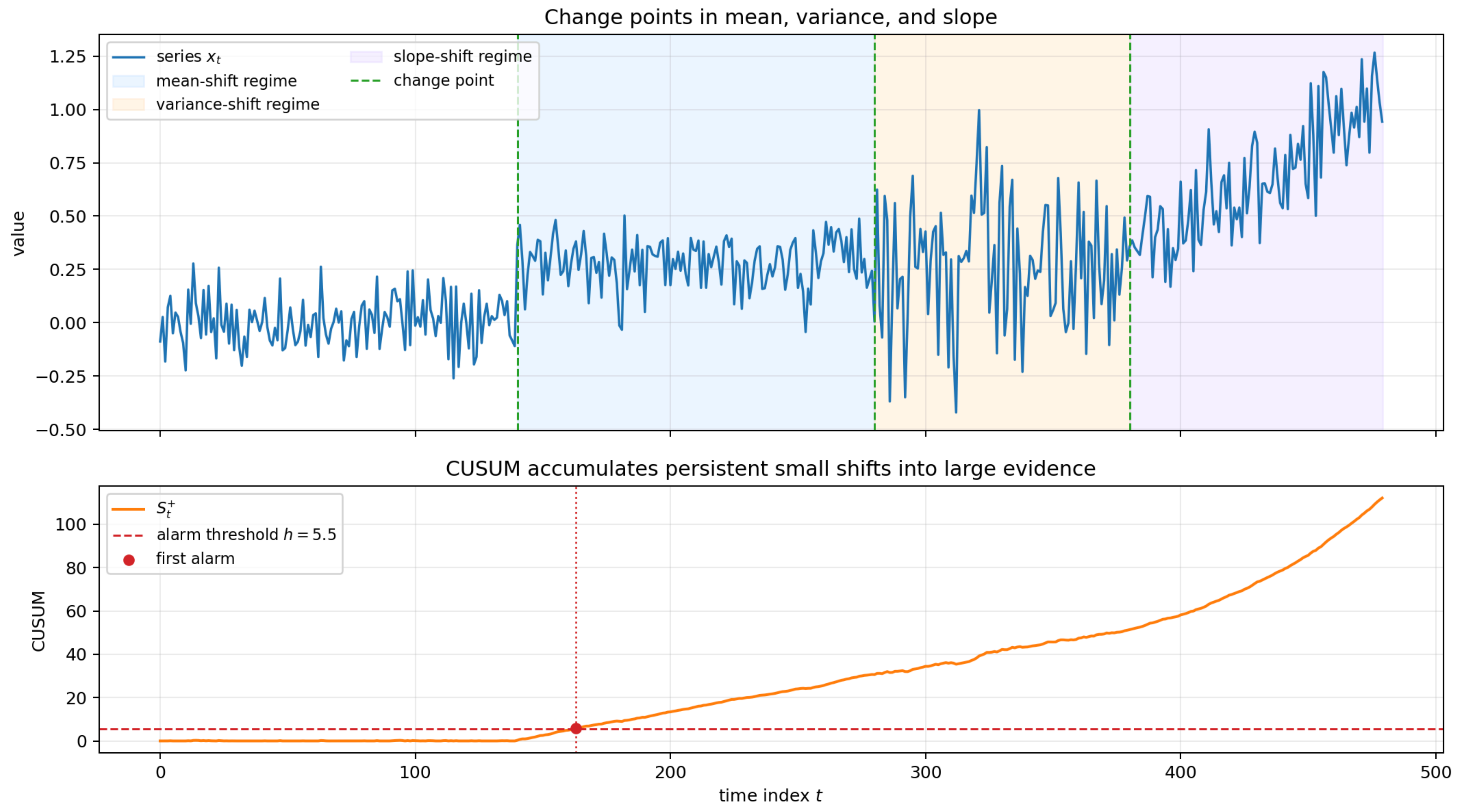
- Decomposition + robust thresholding pattern.
- Process-centric view: this is a statistical pre-processing stage.
- Seasonality caveat: wrong STL period leaks structure into residuals and degrades anomaly separation.
- Benefit: turns contextual anomalies into clearer high-residual events.

STL Decomposition



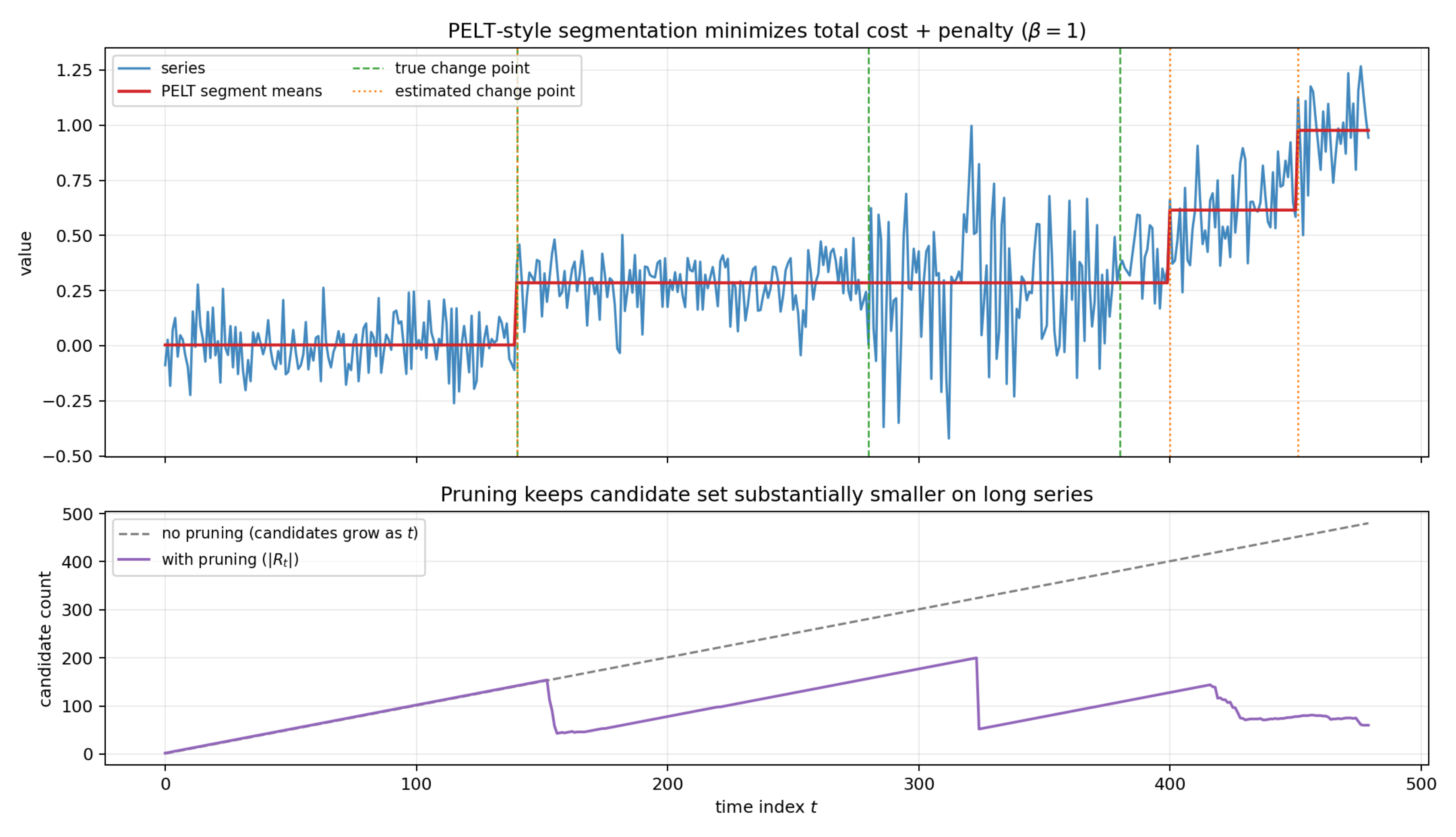
Change-Point Detection: CUSUM and PELT

- **Change point**: timestamp where distribution/regime changes (mean, variance, slope, or combination).
- **CUSUM intuition**:
  - Accumulate signed deviations over time.
  - Persistent small shifts produce growing cumulative evidence.



Change points in mean, variance, and slope
series x_t
mean-shift regime
variance-shift regime
slope-shift regime
change point
value
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
CUSUM accumulates persistent small shifts into large evidence
S_t^+
alarm threshold $h = 5.5$
first alarm
CUSUM
100
80
60
40
20
0
0
100
200
300
400
500
time index t

- **PELT (Pruned Exact Linear Time) intuition**:
  - Segment series by minimizing total segment cost + penalty.
  - Pruning can make long-series search near-linear under common conditions.
- Penalty tradeoff: low penalty over-segments (false alarms), high penalty misses subtle shifts.
- Evaluation fit: regime-shift methods should be judged with event/range metrics, not only point-wise hits.
- Best-fit anomaly types: contextual/collective regime changes rather than isolated spikes.


PELT-style segmentation minimizes total cost + penalty ($\beta = 1$)
series
PELT segment means
true change point
estimated change point
value
Pruning keeps candidate set substantially smaller on long series
no pruning (candidates grow as t)
with pruning ($|R_t|$)
candidate count
time index t

Change-Point Formula Snapshot

- One-sided CUSUM (mean-shift form):
  - $S_t^+ = \max\left(0,\ S_{t-1}^+ + x_t - \mu_0 - k\right)$
  - $S_t^- = \max\left(0,\ S_{t-1}^- + \mu_0 - x_t - k\right)$
  - Alarm when $\max(S_t^+, S_t^-) > h$.

- PELT objective:
    - $$\min_{\tau_1,\ldots,\tau_m} \left[\sum_{i=0}^{m} C\left(x_{(\tau_i+1):\tau_{i+1}}\right) + \beta m \right]$$
    - with $\tau_0 = 0,\ \tau_{m+1} = n.$

Matrix Profile (Definitions)

- **Subsequence length `m`**: window length for motif/discord search.
- **Matrix Profile**: metadata series of nearest-neighbor distances for all length-`m` subsequences (often z-normalized Euclidean).
- **Matrix Profile Index**: nearest-neighbor location for each subsequence (`IMP[i] = j`).
- **Self-join vs Join**: self-join compares subsequences within one series; join compares against another reference series.

- **Motif**: low profile value (highly repeated normal pattern).
- **Discord**: high profile value (most unusual subsequence).
- **Top-k / m-th discord**: useful when anomalies occur in similar groups and nearest-neighbor-only ranking is insufficient.
- Parameter-light behavior: main tuning knob is subsequence length `m`.
- Interpretability: nearest-neighbor match provides a concrete explanation for each motif/discord.

- Parameter sensitivity:
  - Small `m` catches micro-patterns; large `m` catches longer behaviors.
  - `m` should match domain rhythm (cycle fragments, burst durations, drift horizon).
- Retrieval: exclude trivial matches (overlapping windows) when ranking discords.
- Pattern-mining context: SAX and grammar/MDL approaches are alternative symbolic/compression views.

Matrix Profile Formula Snapshot

- Subsequence set for fixed length `m`:
  - $\mathbf{T}_{i,m} = [T_i , T_{i+1} , \ldots , T_{i+m-1}]$
- Matrix Profile value at index `i`:
  - $MP[i] = \min_{j:\ |i-j|>w} d\left(\mathbf{T}_{i,m}, \mathbf{T}_{j,m}\right)$
  - (`w` excludes trivial overlaps)

- Matrix Profile Index:
  - $IMP[i] = \arg \min_{j:\ |i-j|>w} d\left(\mathbf{T}_{i,m}, \mathbf{T}_{j,m}\right)$
- Discord and motif extraction:
  - $$i_{\text{discord}} = \arg\max_i MP[i]$$ $$i_{\text{motif}} = \arg\min_i MP[i]$$

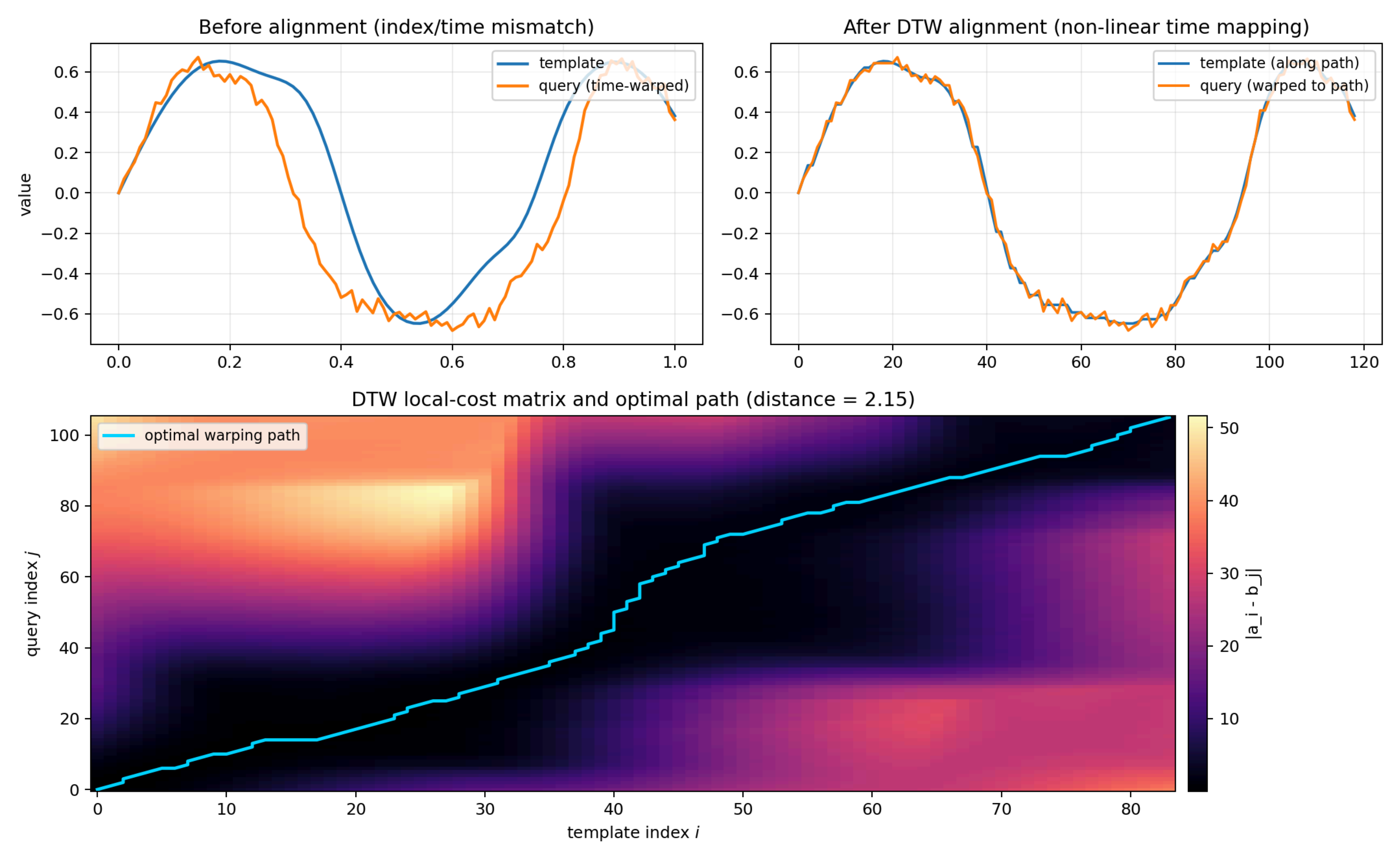
Matrix Profile Motifs and Discords

DTW Primer (Definitions)

- Given two time series $\mathbf{A} = [a_1, \ldots, a_n]$ and $\mathbf{B} = [b_1, \ldots, b_m]$, DTW finds a warping path W that minimizes:

$$\mathrm{DTW}(\mathbf{A}, \mathbf{B}) = \min_{W} \sqrt{\sum_{(i,j)\in W} (a_i - b_j)^2}$$

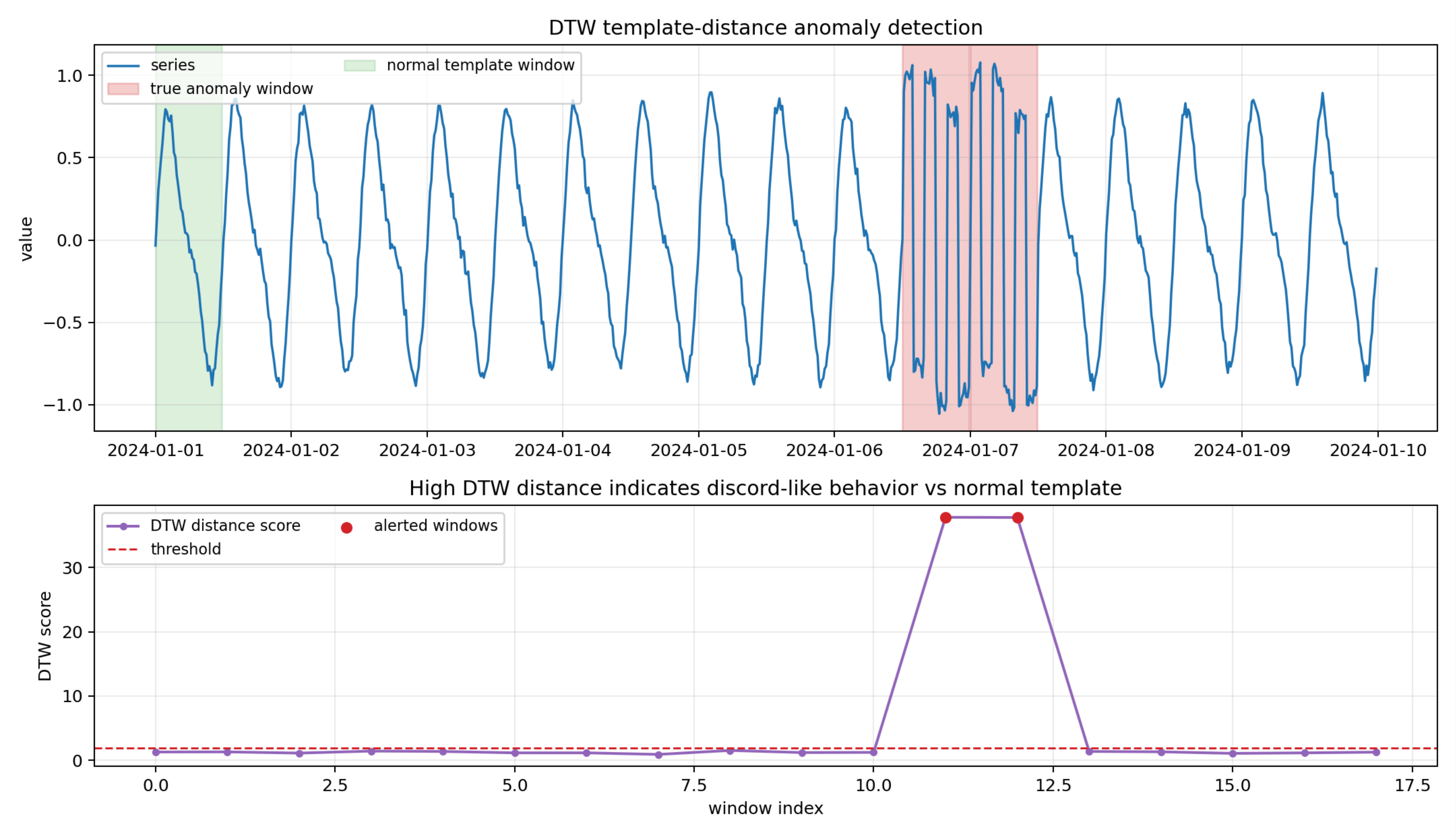
- The path W can move right, up, or diagonally, allowing non-linear alignment.


Before alignment (index/time mismatch)
template
query (time-warped)
value
After DTW alignment (non-linear time mapping)
template (along path)
query (warped to path)
DTW local-cost matrix and optimal path (distance = 2.15)
optimal warping path
query index j
template index i
|a_i - b_j|

- **cDTW**: constrained DTW using a Sakoe-Chiba band to limit unrealistic warping.
- **DDTW**: derivative DTW, comparing local shape changes rather than raw levels.
- **LB_Keogh**: lower bound used to prune expensive DTW comparisons.
- Practical tradeoff: more robust than Euclidean for phase shifts, but computationally heavier without constraints/pruning.
- Distance-family caveat: raw-distance methods may need detrending/normalization because they do not remove seasonality by design.

- Leakage caveat: template banks must be created from past-only normal segments when evaluated in time order.
- Template-distance anomaly scoring:
  - Build normal template(s) from known motifs.
  - Score each new window by minimum DTW distance to template library.
  - High distance implies discord-like behavior under warping-aware similarity.

DTW-Based Anomaly Detection



{fig-alt="Top panel: univariate series with true anomaly windows and

EVT/POT and GPD (Core Concept)

- **EVT** models the behavior of rare extremes, not the full distribution.
- **POT (Peaks Over Threshold)** models excesses above a high threshold t: $Y = X - t \mid X > t$.
- **Pickands-Balkema-de Haan theorem**: for sufficiently high t, the conditional excess distribution converges to a Generalized Pareto Distribution (**GPD**).
- Practical meaning: GPD is the natural tail model for extreme exceedances and the statistical core of POT.

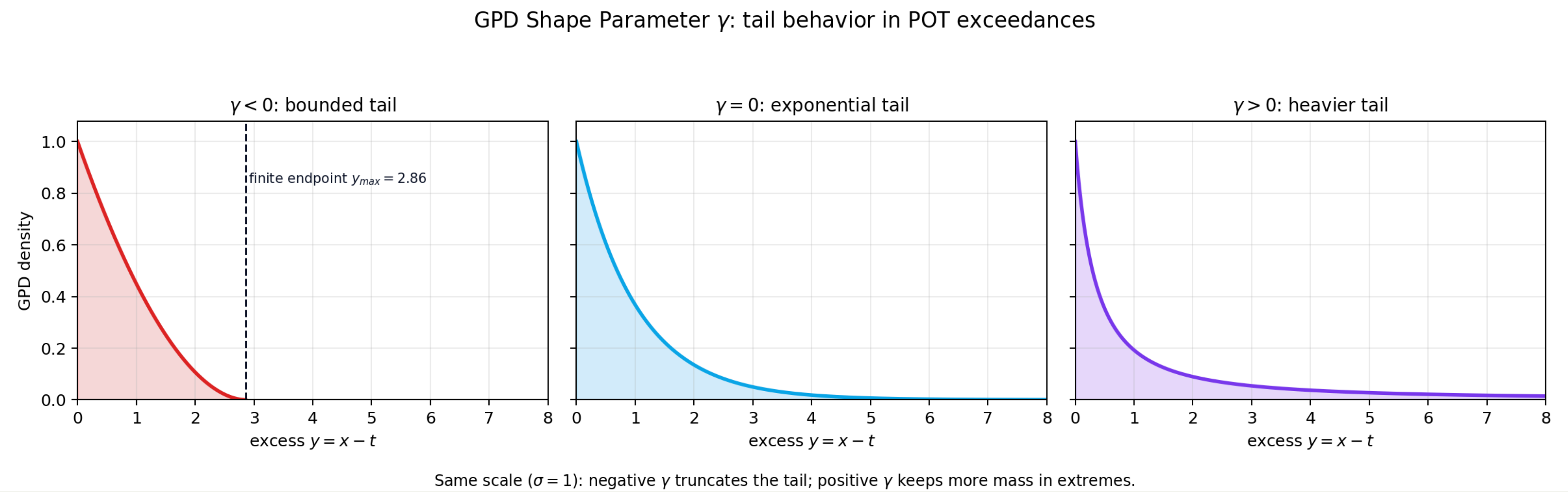
GPD Definition and Parameters

- GPD CDF for excesses $y \geq 0$:

$$G_{\gamma,\sigma}(y) = 1 - \left(1 + \frac{\gamma y}{\sigma}\right)^{-1/\gamma}$$

- **Shape γ (tail index)**:
  - $\gamma = 0$: exponential (light tail)
  - $\gamma > 0$: heavy tail (Pareto-like)
  - $\gamma < 0$: bounded upper tail
- **Scale $\sigma > 0$**: spread of exceedances; depends on threshold choice.
- In POT excess form $Y = X - t$, location parameter is effectively $\mu = 0$.

Visual: GPD Shape Parameter γ



POT Workflow for Anomaly Thresholds

1. Choose high baseline threshold t (often high empirical quantile).
2. Extract exceedances: $Y_i = X_i - t$ for all $X_i > t$.
3. Fit GPD on $\{Y_i\}$ to estimate $\hat{\gamma}, \hat{\sigma}$.
4. Compute extreme decision threshold z_q for target exceedance probability q:

$$z_q \simeq t + \frac{\hat{\sigma}}{\hat{\gamma}}\left[\left(\frac{qn}{N_t}\right)^{-\hat{\gamma}} - 1\right]$$

   where n = calibration size, N_t = number of exceedances.

- Operational rule: flag anomaly if future observation $X > z_q$.

Estimation in Practice (MLE + Grimshaw)

- Fit GPD parameters by **Maximum Likelihood Estimation (MLE)** on exceedances.
- Log-likelihood for exceedances $Y_1, \ldots, Y_{N_t}$: $$ L(,) = -N_t
- (1 +) _{i=1}^{N_t} (1 + Y_i) $$
- Optimization is usually solved with the **Grimshaw trick** for numerical stability under GPD constraints.
- Modelling caveat: threshold t must be high enough for asymptotic validity but low enough to keep enough peaks.

GPD/POT Takeaways

- GPD is the tail model that makes POT principled.
- It turns ad-hoc tail cutoffs into probabilistic thresholds tied to target risk level q.
- This is especially useful when residual/score tails are heavy and static thresholds are unstable.